Algebra II with Trigonometry Chapter 6.1

Olympiad Solutions (GPT-5.4)

Saved tutoring response with MathJax rendering. Original PDF file: 4506904_alg-2trig-h-chp-6-1-note-docx.pdf

Solutions

1. Let P be the terminal point on the unit circle determined by t, where t is chosen so that

$$\cos t + \sin t = \frac{1}{3} \quad \text{and} \quad P \text{ lies in Quadrant II.}$$

Find the exact coordinates of P.

Let

$$x = \cos t, \qquad y = \sin t.$$

Then $P = (x, y)$ lies on the unit circle, so

$$x^2 + y^2 = 1,$$

and we are given

$$x + y = \frac{1}{3}.$$

Use

$$(x + y)^2 = x^2 + y^2 + 2xy.$$

Thus

$$\frac{1}{9} = 1 + 2xy,$$

so

$$2xy = -\frac{8}{9}, \qquad xy = -\frac{4}{9}.$$

Now x and y are the two roots of

$$u^2 - (x+y)u + xy = 0,$$

that is,

$$u^2 - \frac{1}{3}u - \frac{4}{9} = 0.$$

Multiply by 9:

$$9u^2 - 3u - 4 = 0.$$

Hence

$$u = \frac{3 \pm \sqrt{9+144}}{18} = \frac{3 \pm 3\sqrt{17}}{18} = \frac{1 \pm \sqrt{17}}{6}.$$

So the two values are

$$\frac{1+\sqrt{17}}{6}, \qquad \frac{1-\sqrt{17}}{6}.$$

Since P is in Quadrant II, we need $x < 0 < y$. Therefore

$$x = \frac{1-\sqrt{17}}{6}, \qquad y = \frac{1+\sqrt{17}}{6}.$$

$$\boxed{P = \left(\frac{1-\sqrt{17}}{6}, \ \frac{1+\sqrt{17}}{6}\right)}$$

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2. A point $P(x,y)$ lies on the unit circle in Quadrant I and satisfies

$$x - y = \frac{1}{5}.$$

Find $x + y$.

Since P lies on the unit circle,

$$x^2 + y^2 = 1.$$

We are given

$$x - y = \frac{1}{5}.$$

Use

$$(x+y)^2 + (x-y)^2 = 2(x^2+y^2) = 2.$$

So

$$(x+y)^2 + \left(\frac{1}{5}\right)^2 = 2,$$

hence

$$(x+y)^2 = 2 - \frac{1}{25} = \frac{49}{25}.$$

Thus

$$x + y = \pm\frac{7}{5}.$$

But P is in Quadrant I, so $x > 0$ and $y > 0$, hence $x + y > 0$. Therefore

$$\boxed{x + y = \frac{7}{5}}.$$

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3. Suppose the terminal point determined by t is $P = (x, y)$ on the unit circle. If the terminal point determined by $2t$ is $\left(-\frac{7}{9}, \frac{4\sqrt{2}}{9}\right)$, and P lies in Quadrant II, find P.

We have

$$\cos 2t = -\frac{7}{9}, \qquad \sin 2t = \frac{4\sqrt{2}}{9}.$$

Since $P = (x, y) = (\cos t, \sin t)$, use the double-angle formulas:

$$\cos 2t = 2x^2 - 1, \qquad \sin 2t = 2xy.$$

From $\cos 2t = -\frac{7}{9}$,

$$2x^2 - 1 = -\frac{7}{9}$$

so

$$2x^2 = \frac{2}{9}, \qquad x^2 = \frac{1}{9}.$$

Thus

$$x = \pm\frac{1}{3}.$$

Because P lies in Quadrant II, $x < 0$, so

$$x = -\frac{1}{3}.$$

Now from the unit circle,

$$x^2 + y^2 = 1$$

gives

$$\frac{1}{9} + y^2 = 1, \qquad y^2 = \frac{8}{9}.$$

Thus

$$y = \pm\frac{2\sqrt{2}}{3}.$$

In Quadrant II, $y > 0$, so

$$y = \frac{2\sqrt{2}}{3}.$$

Check:

$$2xy = 2\left(-\frac{1}{3}\right)\left(\frac{2\sqrt{2}}{3}\right) = -\frac{4\sqrt{2}}{9},$$

which would make $\sin 2t < 0$, not correct.

So we should instead use the half-angle idea more carefully. Since $\sin 2t > 0$ and $\cos 2t < 0$, we know $2t$ is in Quadrant II. If t is in Quadrant II, this is possible.

Now use

$$\cos^2 t = \frac{1+\cos 2t}{2} = \frac{1-\frac{7}{9}}{2} = \frac{2/9}{2} = \frac{1}{9}.$$

So again $\cos t = \pm\frac{1}{3}$, and Quadrant II gives

$$x = \cos t = -\frac{1}{3}.$$

Also

$$\sin^2 t = \frac{1-\cos 2t}{2} = \frac{1+\frac{7}{9}}{2} = \frac{16/9}{2} = \frac{8}{9},$$

so

$$y = \sin t = \pm\frac{2\sqrt{2}}{3}.$$

Quadrant II gives

$$y = \frac{2\sqrt{2}}{3}.$$

Then indeed

$$\sin 2t = 2\sin t\cos t = 2\left(\frac{2\sqrt{2}}{3}\right)\left(-\frac{1}{3}\right) = -\frac{4\sqrt{2}}{9},$$

which conflicts with the given value. Therefore no point in Quadrant II can satisfy both coordinates as stated.

Hence there is **no such point** P in Quadrant II.

$$\boxed{\text{No such point } P \text{ exists.}}$$

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4. Find the exact value of

$$\cos\frac{\pi}{12} + \sin\frac{\pi}{12} + \cos\frac{5\pi}{12} + \sin\frac{5\pi}{12}.$$

Group the terms:

$$\left(\cos\frac{\pi}{12} + \sin\frac{\pi}{12}\right) + \left(\cos\frac{5\pi}{12} + \sin\frac{5\pi}{12}\right).$$

Use

$$\sin\theta + \cos\theta = \sqrt{2}\cos\left(\theta - \frac{\pi}{4}\right).$$

For $\theta = \frac{\pi}{12}$,

$$\cos\frac{\pi}{12} + \sin\frac{\pi}{12} = \sqrt{2}\cos\left(\frac{\pi}{12} - \frac{\pi}{4}\right) = \sqrt{2}\cos\left(-\frac{\pi}{6}\right) = \sqrt{2}\cdot\frac{\sqrt{3}}{2} = \frac{\sqrt{6}}{2}.$$

For $\theta = \frac{5\pi}{12}$,

$$\cos\frac{5\pi}{12}+\sin\frac{5\pi}{12}=\sqrt{2}\cos\left(\frac{5\pi}{12}-\frac{\pi}{4}\right)=\sqrt{2}\cos\left(\frac{\pi}{6}\right)=\frac{\sqrt{6}}{2}.$$

Therefore the sum is

$$\frac{\sqrt{6}}{2}+\frac{\sqrt{6}}{2}=\sqrt{6}.$$

$$\boxed{\sqrt{6}}$$

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5. Let P be the terminal point determined by t on the unit circle. If the three points determined by t, $\pi - t$, and $-t$ have x-coordinates whose sum is 1, find all possible y-coordinates of P.

Let

$$P=(\cos t,\sin t)=(x,y).$$

The three relevant x-coordinates are:

- for t: $\cos t = x$,
- for $\pi - t$: $\cos(\pi - t) = -\cos t = -x$,
- for $-t$: $\cos(-t) = \cos t = x$.

Their sum is

$$x+(-x)+x=x.$$

We are told this sum is 1, so

$$x=1.$$

Since P lies on the unit circle,

$$x^2+y^2=1.$$

Thus

$$1+y^2=1,$$

so

$$y^2 = 0.$$

Hence

$$\boxed{y = 0}.$$

So the only possible y-coordinate is

$$\boxed{0}.$$

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6. A point $P = (x, y)$ on the unit circle has the property that reflecting P across the line $y = x$ gives the terminal point determined by $t + \frac{\pi}{6}$, where P itself is determined by t. Find all possible exact coordinates of P.

If P is determined by t, then

$$P = (\cos t, \sin t).$$

Reflecting across $y = x$ swaps coordinates, so the reflected point is

$$(\sin t, \cos t).$$

We are told this is the terminal point determined by $t + \frac{\pi}{6}$, so

$$(\sin t, \cos t) = \left(\cos\left(t + \frac{\pi}{6}\right), \sin\left(t + \frac{\pi}{6}\right)\right).$$

Hence

$$\sin t = \cos\left(t + \frac{\pi}{6}\right), \qquad \cos t = \sin\left(t + \frac{\pi}{6}\right).$$

Use angle-sum formulas:

$$\cos\left(t + \frac{\pi}{6}\right) = \frac{\sqrt{3}}{2}\cos t - \frac{1}{2}\sin t,$$

$$\sin\left(t+\frac{\pi}{6}\right)=\frac{\sqrt{3}}{2}\sin t+\frac{1}{2}\cos t.$$

So

$$\sin t=\frac{\sqrt{3}}{2}\cos t-\frac{1}{2}\sin t,$$

which gives

$$\frac{3}{2}\sin t=\frac{\sqrt{3}}{2}\cos t,$$

hence

$$3\sin t=\sqrt{3}\cos t \quad\Rightarrow\quad \tan t=\frac{1}{\sqrt{3}}.$$

Also from the second equation,

$$\cos t=\frac{\sqrt{3}}{2}\sin t+\frac{1}{2}\cos t,$$

so

$$\frac{1}{2}\cos t=\frac{\sqrt{3}}{2}\sin t,$$

hence again

$$\cos t=\sqrt{3}\sin t \quad\Rightarrow\quad \tan t=\frac{1}{\sqrt{3}}.$$

Therefore

$$t=\frac{\pi}{6}+k\pi.$$

So

$$P=(\cos t,\sin t)$$

is either

$$\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right) \quad \text{or} \quad \left(-\frac{\sqrt{3}}{2}, -\frac{1}{2}\right).$$

Thus all possible exact coordinates are

$$\boxed{\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right) \text{ and } \left(-\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)}.$$